```
function [sig2_est,Nref,final_althyp_index] =
ps_network(Nifgs,Npsc,Npsc_selections,Nref,max_arc_length,psc_model,final_model,
Btemp,Bdop,std_param,breakpoint,breakpoint2,network_method,Ncon,Nparts,ens_coh_t
hreshold,varfac_threshold,ref_cn)

% Network estimation
%
% Input:    - Nifgs              number of interferograms
%           - Npsc               number of psc's
%           - Npsc_selections    number of psc_selections
%           - Nref               number of reference points
%           - max_arc_length     maximum arc length
%           - psc_model          vector with model indices to be
%                                evaluated (see help model_definitions
%                                for more information)
%           - final_model        model used for unwrapped data
%           - Btemp              temporal baselines [year]
%           - Bdop               Doppler baselines [Hz]
%           - std_param          standard deviations for pseudo-
%                                observations
%           - breakpoint         breakpoint in case of double
%                                linear model
%           - breakpoint2        second breakpoint, should be larger
%                                than first breakpoint
%           - network_method     network identifier, 'delaunay' or
%                                'spider'
%           - Ncon               minimum number of connections (arcs)
%                                to a psc (network 2 only)
%           - Nparts             number of partitions of a full cycle
%                                to which the arcs are divided
%                                (network 2 only)
%           - ens_coh_threshold  ensemble coherence threshold
%           - varfac_threshold   variance factor threshold
%           - ref_cn             coordinates reference point
%
% Output:   - sig2_est           variance components
%           - Nref               number of reference points
%           - final_althyp_index althyp_index used for unwrapped data
%
% ------------------------------------------------------------------------
% File.............: ps_network.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
% Authors.........: Freek van Leijen
%                   Delft Institute of Earth Observation and Space Systems
%                   Delft University of Technology
% ------------------------------------------------------------------------
%
% This software is developed by Delft University of Technology and is
% intended for scientific use only. Applications for commercial use are
% prohibited.
%
% Copyright (c) 2004-2009 Delft University of Technology, The Netherlands
%
% Change log
% v1.7.2.9, Freek van Leijen
% - flag for removed points, unwrap2_istep
%



% ------------------------------------------------------------------------
% Initialize
% ------------------------------------------------------------------------
```

```
global ps_method unwrap_istep unwrap2_istep results_id visible_plots


% ------------------------------------------------------------------------
% 4.1
% Variance Component Estimation (VCE)
% ------------------------------------------------------------------------

fprintf(1,'\n');
fprintf(1,'Variance component estimation ....\n');

removed_points_flag_array = ones(Npsc_selections,1);
unwrap2_istep = 0;

while sum(removed_points_flag_array,1)~=0 & unwrap2_istep<=1
  %maximum one iteration

  unwrap2_istep = unwrap2_istep+1;% to account for removed psc

  sig2_est = ps_vce(Npsc,...
                Nifgs,...
                max_arc_length,...
                psc_model,...
                Btemp,...
                Bdop,...
                std_param,...
                breakpoint,...
                breakpoint2);



  for z = 1:Npsc_selections

    network_flag = 0;
    unwrap_istep = 0;

    while network_flag==0

      unwrap_istep = unwrap_istep+1;% to account for separate networks

      % ------------------------------------------------------------------------
      % 4.2
      % Form network between initial ps
      % ------------------------------------------------------------------------

      fprintf(1,'\n');
      fprintf(1,'Forming network ....\n');

      Narcs = ps_form_network(Npsc,...
                      Npsc_selections,...
                      Nifgs,...
                      max_arc_length,...
                      network_method,...
                      Ncon,...
                      Nparts,...
                      z);




      % ------------------------------------------------------------------------
      % 4.3
      % Ps estimation
      % ------------------------------------------------------------------------
```

```
      fprintf(1,'\n');
      fprintf(1,['Ps estimation by ' ps_method ' ....\n']);

      ps_estimation(Btemp,...
                Bdop,...
                psc_model,...
                Nifgs,...
                Narcs,...
                Npsc_selections,...
                std_param,...
                sig2_est,...
                breakpoint,...
                breakpoint2,...
                z);



      % ------------------------------------------------------------------------
      % 4.4
      % Spatial unwrapping
      % ------------------------------------------------------------------------

      fprintf(1,'\n');
      fprintf(1,'Spatial unwrapping ....\n');

      [Nref,...
       final_althyp_index,...
       network_flag,...
       removed_points_flag_array(z)] = ps_spatial_unwrapping(...
                                           ens_coh_threshold,...
                                  varfac_threshold,...
                                  ref_cn,...
                                  Nifgs,...
                                  Npsc,...
                                  Npsc_selections,...
                                  Nref,...
                                  Narcs,...
                                  Btemp,...
                                  Bdop,...
                                  final_model,...
                                  std_param,...
                                  sig2_est,...
                                  breakpoint,...
                                  breakpoint2,...
                                  network_flag,...
                                  z);

    end
  end
end
```